

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**First/Second Semester of B.Tech (ME /CL/ EE) Examination****April - May 2018****IT141/IT101 Fundamentals of Computer Programming****Date: 09/05/2018, Wednesday****Time: 1.30 p.m. To 4.30 p.m.****Maximum Marks: 70****Instructions:**

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is not allowed.

SECTION – I**Q - 1 Answer the following questions. [11]**

1. Convert the binary number 110101101110 to its equivalent Octal number. 01
2. The _____ header file contains mathematical functions. 01
3. The _____ operator returns the number of bytes the operand occupies. 01
4. 'C' is Robust programming language – Justify the Statement 01
5. What is the significance of comma operator in 'C'? 01
6. Following Statement in 'C' is Valid or Not. Justify the Answer. 01
`#define price of item = $ 50 ;`
7. Which are the characteristics that make 'C' language much popular? 02
8. State True or False. 03
 a) goto is keyword that provide conditional branching in 'C' programming .
 b) nested if...else is not permitted in 'C' Programming.
 c) The default case is compulsory required in switch statement.

Q – 2 Answer the following questions. [12]

1. What is flowchart? List the symbols that are used to draw the flowchart.
2. Explain else if ladder with suitable example.
3. List the name of operators in 'C'. Explain any one operator with suitable example.

OR**Q – 2 Answer the following questions. [12]**

1. Write an algorithm to find the reverse number of given integer number.
2. Explain the character test functions isalpha() and isdigit() with suitable example .
3. Explain switch statement with suitable example.

Q - 3 Answer the following questions. [12]

1. Write a program to accept two numbers from user and interchange value of two variables using following criteria.
 - a) Using third variable
 - b) Without using third variable
2. Write a program to read an integer value from user and check whether the entered number is odd or even.

OR**Q - 3 Answer the following questions. [12]**

1. Write a program to print 1 to N numbers and sum of them using while, do...while loop.
2. Write a program to print a following triangle using nested *for* loop.

```

1
2 2
3 3 3
4 4 4 4
5 5 5 5 5

```

SECTION – II**Q - 4 Answer the question below. [11]**

1. String constants should be enclosed between _____. 01
2. List the advantage and limitation of an Array. 02
3. State True or False. 02
 - a) If return type for a function is not specified, it defaults to int.
 - b) A structure cannot be nested in a structure.
4. Fill in the blanks with most appropriate words from below options. 03
 (strlen(), case, prototype, strcmp(), dot, implementation)
 - i. _____ function compares two strings and returns 0 if both strings are identical.
 - ii. In a structure, member operator is also known as _____ operator.
 - iii. Function declaration is also known as function _____.
5. Classify the following variable names are valid or invalid. 03
 - a) IPL_2018 b) 13saath c) List_052 d) namo e) 4th May f) static

Q – 5 Answer the following questions.**[12]**

1. Write a program to check whether the entered string is palindrome or not.
2. Write a program to find the factorial of given number using recursive function.
3. What is the use of structure in 'C'? Define the structure. Explain how to access the structure members?

OR**Q – 5 Answer the following questions.****[12]**

1. What is a string? Explain the use of `strcat()`, `strcpy()`, `strrev()`.
2. Explain the following terms with respect to user-defined functions
 - i. function definition
 - ii. function declaration
3. Define structure named book with members: `int book_id`, `price`, `char title[10]`. Write a C program that enters data for 3 students using structure and print them in proper format.

Q – 6 Answer the following questions.**[12]****a) Attempt any two.****10**

1. Describe the basic structure of a C program in detail.
2. List out the categories of functions in C. Explain any one category with example.
3. Explain While and do...while loop with suitable example.

b) What will be output of following C code?**02**

1) <pre>#include <stdio.h> int main() { int a = 10, b = 20; if(a=b) { printf("Easy"); } else { printf("Hard"); } return 0; }</pre>	2) <pre>#include <stdio.h> void main() { int i,j,k,count; count=0; for(i=0;i<5;i++) { for(j=0;j<5;j++) { count++ ; } } printf("%d",count); }</pre>
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